

AYUSH SHRIVASTAVA

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SUMMARY

Business Intelligence Analyst and Data Engineer with 4+ years of experience designing scalable data pipelines, predictive models, and interactive dashboards using Python, SQL, and cloud-native tools (AWS, GCP). Proven track record of transforming raw datasets into actionable insights to support supply chain optimization, pricing strategies, and customer segmentation. Skilled in real-time data processing, anomaly detection, and KPI tracking using tools like Spark Streaming, Redshift, BigQuery, and Power BI.

SKILLS:

- **Certifications:** Databricks Data Analyst Associate, Azure AI Fundamentals, Data Science using Python & R.
- **Programming Language:** Python, R, SAS, MATLAB.
- **Data Analysis Tools:** NumPy, Pandas, Matplotlib, SciPy, OpenCV, scikit-learn, Snowflake.
- **Database Management:** SQL (MySQL, PostgreSQL, Redshift), NoSQL (MongoDB), Hadoop, Spark, Oracle Database.
- **Visualization Software:** Tableau, Quicksight, Power BI, Looker, Microsoft Excel.
- **Cloud & Tools:** AWS (EC2, S3, RDS), Azure, Google Cloud Platform, Docker, Kubernetes, Ansible, Terraform, Kafka.
- **Data science & Analytics:** Data Analytics, Data Visualization, Data Modeling, Ad Hoc Analysis, Forecasting Models.

EXPERIENCE:

Business Intelligence Engineer – Kroger Co., Ohio

Jun 2024 - Current

- Developed predictive analytics solutions using **Python**, **SQL**, and **Redshift** to forecast demand across 2,000+ retail locations, optimizing inventory turnover and reducing perishable waste by **11%**. Enabled cross-functional teams to leverage insights for demand forecasting and supply chain efficiency.
- Developed self-service dashboards and automated anomaly detection audits in **Quicksight** and **Power BI**, proactively identifying inventory stockout trends and reducing fulfillment defects by **6%**.
- Established and automated scalable processes using **AWS tools** (EC2, S3, RDS) to optimize transportation routes, improving operational efficiency by **15%** and reducing delivery bottlenecks for high-demand regions.
- Leveraged **clustering** and **regression** techniques to analyze loyalty card data and identify customer segments, which informed personalized promotions and increased repeat purchases by **7%**.
- Collaborated with category managers, marketing, and operations teams to define and track **KPIs** (e.g., basket size, out-of-stock rate, promotion uplift), improving strategic decision-making and store performance reporting.
- Conducted causal analyses of stockout patterns using **Apache Spark Streaming**, integrating real-time sales and inventory data to identify root causes and trigger automated replenishment, reducing stockouts by **8%** across high-demand categories.

Data Engineer – Cognizant Technology Solutions Ltd., India

May 2021 - Aug 2023

- Architected data models for e-commerce platforms, enabling real-time analysis of customer delivery metrics using **SQL**, improving predictability of supply chain operations by **12%**.
- Created **7** interactive **Tableau** and **Power BI** dashboards to track customer satisfaction KPIs (e.g., NPS, order accuracy). Provided actionable insights that optimized customer experience strategies and improved SLA adherence by **10%**.
- Built scalable data pipelines on **Google Cloud Platform** using **BigQuery**, **Cloud Functions**, and **Cloud Composer** to automate ETL workflows across marketing and logistics datasets, reducing manual intervention by **25%**.
- Facilitated stakeholder workshops to communicate operational trends identified through statistical methods (e.g., t-tests, Chi-squared analysis) and proposed strategies that reduced costs by **10%**.
- Designed **SQL & Python**-based audit pipelines to detect data anomalies and ensure data integrity, reducing processing errors by **18%**. Improved data governance and compliance with internal security policies.

Junior Data Engineer – Cognizant Technology Solutions Ltd., India

Aug 2020 - Apr 2021

- Assisted in automating data collection workflows, improving the timeliness and accuracy of reports delivered to stakeholders.
- Optimized inventory management processes by implementing **R-based analytics** to forecast motorcycle parts and accessories needs, reducing overstock by **10%** and aligning inventory turnover with customer purchase trends.

EDUCATION:

University of Dayton, Ohio

Sep 2023 – Jun 2025

Master of Science in Business Analytics

GPA - 3.50

Ramdeobaba College of Engineering and Management, India

Jun 2017 - Jun 2021

Bachelor of Engineering in Electronics Design Technology

GPA - 3.70

ACADEMIC PROJECTS:

Predictive Analytics for Colorectal Cancer Outcomes (SEER) – Developed for National Cancer Institute

- Developed a scalable machine learning pipeline using Python, Pandas, and Scikit-learn to process SEER data (~800K records, 149 features), incorporating robust imputation, encoding, and correlation-based feature filtering.
- Engineered a preprocessing module with integrated NLP to extract insights from physician notes, enhancing model inputs with structured and unstructured data.
- Trained multiple classification models (Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost), applying cross-validation, GridSearchCV, and threshold optimization to maximize AUC, precision, and recall — achieving a top AUC of 0.89.
- Deployed the solution via a Streamlit dashboard, integrating Elasticsearch to enable real-time search and filtering of patient metadata alongside model predictions; version-controlled with Git for reproducibility and stakeholder collaboration.